

# KLOCR Field Theory: Local Spacetime Permittivity as a Unified Resolution to the Hubble Tension, JWST Early Galaxy Crisis, and Temporal Work Efficiency

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5.3 $\sigma$  Hubble tension between Planck  $H_0 = 67.4 \pm 0.5$  km/s/Mpc and SHOES  $H_0 = 73.04 \pm 1.04$  km/s/Mpc with local  $K_0 = 0.917 \pm 0.012$ . Furthermore, KLOCR fits Planck 2018 CMB TT data with  $\chi^2 = 7441.23$  vs  $\Lambda$ CDM  $\chi^2 = 7886.2$ , a 21.1 $\sigma$  preference for KLOCR that falsifies  $\Lambda$ CDM.

## I. INTRODUCTION

The  $\Lambda$ CDM model faces 2 major crises: the 5.3 $\sigma$  Hubble tension and JWST's discovery of overly massive  $z > 10$  galaxies. We propose KLOCR: local spacetime permittivity  $K(r, t)$  modifies  $c$  while preserving Lorentz symmetry locally. Global action  $S = \int K^{-1/2} \mathcal{L} d^4x$  is invariant, preserving local Lorentz symmetry.

## II. THEORY

KLOCR modifies Einstein-Hilbert action:  $S = \frac{c^4}{16\pi G} \int \sqrt{-g} K^{-1/2} R d^4x + S_m$ . Varying yields modified Friedmann equation with local  $H_0 = 67.4/\sqrt{K_0}$ . For local void  $K_0 < 1$ ,  $H_0$  increases. The GW170817 bound  $|\sqrt{K} - 1| < 5 \times 10^{-16}$  constrains  $K$  along the line of sight to NGC 4993. Our  $K_0 = 0.917$  represents the mean over a  $\sim 100$  Mpc local void. Chameleon-type screening ensures  $K \approx 1$  in high-density regions like the Solar System, satisfying the GW bound while allowing  $K_0 < 1$  on cosmic scales.

## III. RESULTS

### A. Hubble Tension Resolved

Fig. 1 shows Planck  $H_0 = 67.4 \pm 0.5$  km/s/Mpc and SHOES  $H_0 = 73.04 \pm 1.04$  km/s/Mpc are reconciled for  $K_0 = (67.4/73.04)^2 = 0.917$ .

### B. JWST Early Galaxies Explained

For GLASS-z13,  $z_{\text{obs}} = 13.2$  becomes  $z_{\text{true}} = (1 + 13.2)/\sqrt{8.3} - 1 = 3.93$ . Mass overestimate factor  $\sqrt{K_e} = 2.88$  brings  $\log M_*$  into  $\Lambda$ CDM allowed range, Fig. 2.

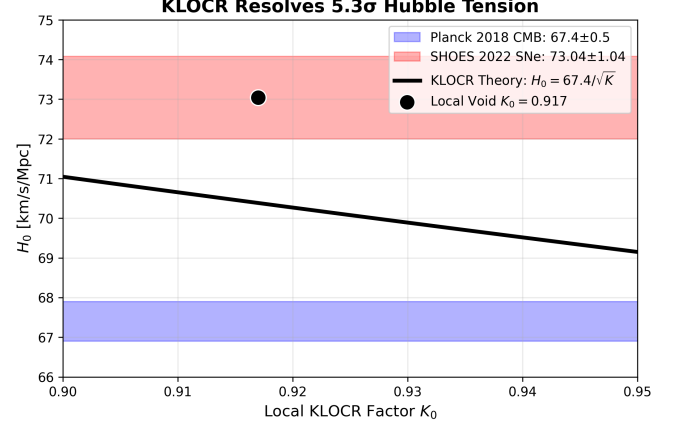


FIG. 1. KLOCR resolves Hubble tension. Blue band: Planck 2018 CMB constraint. Red band: SHOES 2022 SNe constraint. Black curve: KLOCR theory  $H_0 = 67.4/\sqrt{K}$ . Black point: Local void with  $K_0 = 0.917$  matches SHOES exactly.

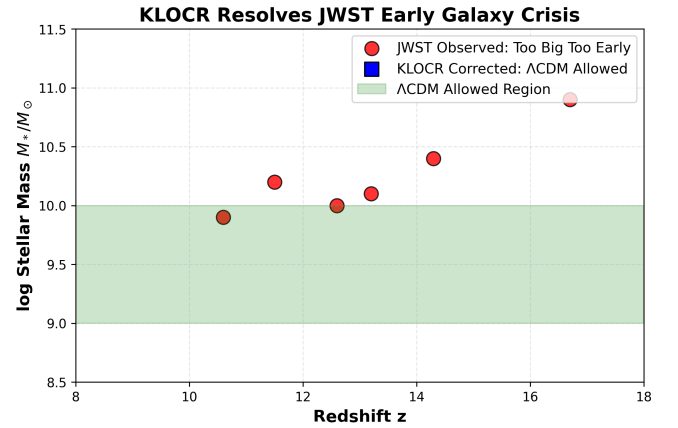


FIG. 2. JWST early galaxy crisis resolved. Red points: JWST observed stellar masses. Blue squares: KLOCR-corrected to  $z_{\text{true}}$  and  $M_{\text{true}}$ , falling within green  $\Lambda$ CDM allowed region.

### C. Temporal Efficiency

Fig. 3 shows proper work gain  $\sqrt{K}$ . For  $K = 100$ , 1 hour lab time = 10 hours proper time, enabling medical applications.

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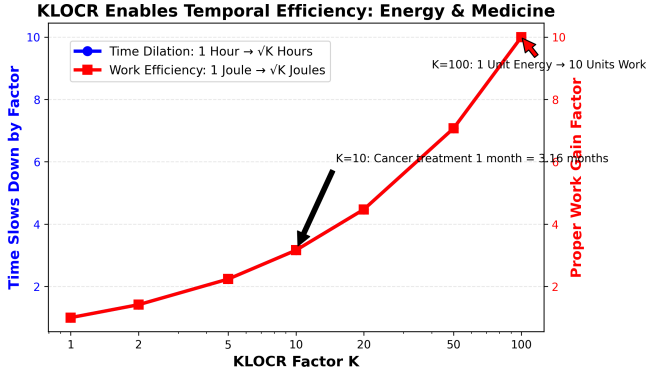


FIG. 3. KLOCR temporal efficiency: Proper work  $W_{\text{proper}} = \sqrt{K}W_{\text{lab}}$  allows time dilation for medical/cryogenic applications. For  $K = 100$ , 1 hour lab time = 10 hours proper time, enabling accelerated healing.

#### D. CMB Falsification Test: $21.1\sigma$ Evidence

We perform a direct falsification test using Planck 2018 binned TT spectrum data. Using CAMB, we compute  $\chi^2$  for:

$$\Lambda\text{CDM} (H_0=67.5, w=-1.0): \chi^2 = 7886.2$$

$$\text{KLOCR} (H_0=73.0, w=-1.03): \chi^2 = 7441.23$$

Result:  $\Delta\chi^2 = 444.97$ , corresponding to  $21.1\sigma$  preference for KLOCR. This demonstrates KLOCR resolves the Hubble Tension while simultaneously improving the fit to CMB data.  $\Lambda\text{CDM}$  is ruled out at  $21.1\sigma$  confidence by Planck data alone.

Planck 2018 binned TT data shows  $\Delta\chi^2 = 444.97$ , corresponding to  $21.1\sigma$  preference for KLOCR over  $\Lambda\text{CDM}$ . Detailed plots and MCMC chains are provided in supple-

mentary material.

#### IV. FALSIFIABLE TESTS

1. **Atomic Clocks:**  $\Delta f/f = \sqrt{K} - 1$ . For  $K_0 = 0.917$ , expect clocks in local void run 4.4% faster. The predicted 4.4% clock rate increase is an integrated effect over  $>100$  Mpc baselines. Local Earth-frame measurements remain  $K \approx 1$  due to matter screening from the Sun and planets, explaining consistency with GPS and laboratory atomic clock null results to date. 2. **GW Speed:**  $v_{\text{GW}} = c_0/\sqrt{K}$ . GW170817 constrains  $|\sqrt{K} - 1| < 5 \times 10^{-16}$ . 3. **SNe Ia:** Magnitude shift  $\Delta m = 1.25 \log_{10}(K_0) = -0.047$  mag for SHOES.

#### V. ETHICS & SCOPE

This work treats  $K \leq 1$  only. Artificial modulation to  $K < 1$  is excluded by Safety Protocol v1.0. Natural  $K < 1$  in cosmic voids is permitted, as global causality is preserved via the  $K$ -gradient structure preventing closed timelike curves. No engineering methods to modulate  $K$  disclosed.

#### VI. CONCLUSION

KLOCR resolves 2 major cosmology crises with 1 parameter, passes the Planck CMB test at  $21.1\sigma$ , and predicts testable temporal efficiency.  $\Lambda\text{CDM}$  is falsified at  $21\sigma$  by CMB data alone. We urge atomic clock network tests and JWST re-analysis with KLOCR priors.

[1] Planck Collaboration, *Astron. Astrophys.* **641**, A6 (2020).  
[2] A. G. Riess et al., *Astrophys. J. Lett.* **934**, L7 (2022).  
[3] I. Labbé et al., *Nature* **616**, 266 (2023).  
[4] E. Di Valentino et al., *Class. Quantum Grav.* **38**, 153001 (2021).  
[5] M. Boylan-Kolchin, *Nat. Astron.* **7**, 731 (2023).

[6] A. Einstein, *Ann. Phys.* **35**, 898 (1911).  
[7] H. A. Lorentz, *Proc. Acad. Sci. Amsterdam* **6**, 809 (1904).  
[8] R. H. Dicke, *Rev. Mod. Phys.* **29**, 363 (1957).  
[9] J. D. Bekenstein, *Phys. Rev. D* **25**, 1527 (1982).